



Secular coefficients mini-course

Talk 4

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Recall Keating and Snaith gave the prediction

$$M_k(T) := \frac{1}{T} \int_0^T \left| \zeta \left(\frac{1}{2} + it \right) \right|^{2k} dt \sim a_k g_k (\log T)^{k^2}$$

for $g_k = \prod_{j=0}^{k-1} \frac{j!}{(j+k)!}$. Known only for $k = 1, 2$; for $k = 3, 4$ consistent with predictions obtained by other methods.

This was motivated by an analogous result for $U(N)$:

Theorem 1 (Keating–Snaith)

We have

$$\mathbb{E} |\det(I - U_N)|^{2k} = \prod_{j=1}^N \frac{\Gamma(j)\Gamma(j+2k)}{(\Gamma(j+k))^2} \sim g_k N^{k^2}.$$

How can one approach moments of ζ ? One uses the approximation

$$\zeta(1/2 + it) \approx \sum_{n \leq T} n^{-1/2 - it}$$

which is quite good when $T \asymp t$.

Then observe that the functions $(n^{-it})_{n \leq T}$ are essentially orthogonal in $L^2([0, T])$, so

$$\begin{aligned} \frac{1}{T} \int_0^T |\zeta(1/2 + it)|^2 dt &\approx \frac{1}{T} \int_0^T \left| \sum_{n \leq T} n^{-1/2 - it} \right|^2 dt \\ &\approx \frac{1}{T} \int_0^T \sum_{n \leq T} |n^{-1/2 - it}|^2 dt = \sum_{n \leq T} \frac{1}{n} \sim \log T. \end{aligned}$$

This approach does not extend to higher moments because

$$\zeta(1/2 + it) \approx \sum_{n \leq T} n^{-1/2 - it}$$

for $T \asymp t$ implies

$$\zeta(1/2 + it)^k \approx \sum_{n \leq T^k} a(n, k) n^{-1/2 - it}$$

for certain coefficients $a(n, k)$, but it is no longer true that $(n^{-it})_{n \leq T^k}$ are orthogonal in $L^2([0, T])$ once $k \geq 2$.

Conrey–Gamburd looked at the following limit, which they called a pseudomoment:

$$PM_k(x) := \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \left| \sum_{n \leq x} \frac{1}{n^{1/2+it}} \right|^{2k}.$$

Because the summand only depends on p^{it} for $p \leq x$, and $(p^{it})_{p \leq x}$ for t uniform in $[0, T]$ tends to $(\alpha(p))_{p \leq x}$, then

$$PM_k(x) = \mathbb{E} \left| \sum_{n \leq x} \frac{\alpha(n)}{n^{1/2}} \right|^{2k},$$

i.e. moments of Steinhaus.

By expanding the $2k$ th power, and using $\mathbb{E}[\alpha(n)\overline{\alpha(m)}] = \delta_{n,m}$, we obtain

$$PM_k(x) = \sum_{\substack{n_1, \dots, n_k \leq x \\ m_1, \dots, m_k \leq x \\ \prod n_i = \prod m_j}} \frac{1}{\sqrt{n_1 \cdots n_k}} \frac{1}{\sqrt{m_1 \cdots m_k}}.$$

Theorem 2 (Conrey–Gamburd, 2003)

We have

$$PM_k(x) \sim a_k g'_k (\log x)^{k^2}$$

for

$$g'_k = \text{vol}(\mathcal{P}_k), \quad \mathcal{P}_k = \{(x_{i,j}) \in \mathbb{R}_{\geq 0}^{k^2} : \sum_{i=1}^k x_{i,j} \leq 1, \sum_{j=1}^k x_{i,j} \leq 1\}.$$

The proof of Conrey–Gamburd is complex-analytic: define

$$F(s_1, \dots, s_k, t_1, \dots, t_k) = \sum_{\substack{n_1, \dots, n_k \\ m_1, \dots, m_k \geq 1 \\ \prod n_i = \prod m_j}} \prod_{i=1}^k n_i^{-s_i} \prod_{j=1}^k m_j^{-t_j}$$

and apply Perron's formula to each of the variables of F .

Motivation for Conrey–Gamburd (I)

The expression

$$\sum_{n \leq x} \frac{\alpha(n)}{\sqrt{n}}$$

is analogous to the following sum in finite fields:

$$\sum_{\deg f \leq n} \frac{\alpha_q(f)}{|f|^{1/2}},$$

where α_q is Steinhaus on polynomials. This is just the sum of the first n coefficients of $\mathcal{L}(u/\sqrt{q}, \alpha_q)$, so it is also analogous to

$$\sum_{n \leq N} c_n.$$

Motivation for Conrey–Gamburd (II)

It can be shown that

$$\mathbb{E} \left| \sum_{n \leq N} c_n \right|^{2k}$$

is precisely the number of $k \times k$ arrays of nonnegative integers, where each row sum and each column sum does not exceed N ; this grows like $g'_k N^{k^2}$. (Diaconis–Gamburd)

Is there a more transparent link between the two results? Yes!

Moments of c_n (I)

To prove the aforementioned formula for moments of $\sum_{n \leq N} c_n$, it suffices to prove

$$\mathbb{E}\left[\prod_{i=1}^k c_{n_i} \prod_{j=1}^k \overline{c_{m_j}}\right]$$

equals the number of $k \times k$ arrays of nonnegative integers, where the i th row sums to n_i , and j th column sums to m_j .

This is the coefficient of $\prod_{i=1}^k u_i^{n_i} \prod_{j=1}^k v_j^{m_j}$ in

$$\mathbb{E}\left[\prod_{i=1}^k F(u_i) \prod_{j=1}^k \overline{F}(v_j)\right].$$

Moments of c_n (II)

The expectation $\mathbb{E}[\prod_{i=1}^k F(u_i) \prod_{j=1}^k \overline{F}(v_j)]$ simplifies as

$$\begin{aligned}\mathbb{E}[\prod_{i=1}^k F(u_i) \prod_{j=1}^k \overline{F}(v_j)] &= \prod_{n=1}^{\infty} \mathbb{E} \exp\left(-\frac{N_n}{\sqrt{n}} \sum_{i=1}^k u_i^n - \frac{\overline{N}_n}{\sqrt{n}} \sum_{j=1}^k v_j^n\right) \\ &= \prod_{n=1}^{\infty} \exp\left(\frac{1}{n} \left(\sum_{i=1}^k u_i^n\right) \left(\sum_{j=1}^k v_j^n\right)\right) \\ &= \prod_{1 \leq i, j \leq k} \frac{1}{1 - u_i v_j}.\end{aligned}$$

Moments of c_n (III)

The coefficient of $\prod u_i^{n_i} \prod v_j^{m_j}$ in

$$\prod_{1 \leq i, j \leq k} \frac{1}{1 - u_i v_j}$$

is computed by expanding

$$\frac{1}{1 - u_i v_j} = \sum_{r_{i,j} \geq 0} (u_i v_j)^{r_{i,j}}.$$

The required coefficient is the number of solutions $r_{i,j} \geq 0$ to

$$\sum_j r_{i,j} = n_i \quad (i = 1, \dots, k), \quad \sum_i r_{i,j} = m_j \quad (j = 1, \dots, k).$$

Each solution corresponds to an array $(r_{i,j})_{1 \leq i, j \leq k}$.

Conrey–Gamburd used complex analysis to prove that

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \left| \sum_{n \leq x} \frac{1}{n^{1/2+it}} \right|^{2k} = \sum_{\substack{n_1, \dots, n_k \leq x \\ m_1, \dots, m_k \leq x \\ \prod n_i = \prod m_j}} \frac{1}{\sqrt{n_1 \cdots n_k}} \frac{1}{\sqrt{m_1 \cdots m_k}}$$

grows like $a_k g'_k (\log x)^{k^2}$ for

$$g'_k = \text{vol}(\mathcal{P}_k), \quad \mathcal{P}_k = \{(x_{i,j}) \in \mathbb{R}_{\geq 0}^{k^2} : \sum_{i=1}^k x_{i,j} \leq 1, \sum_{j=1}^k x_{i,j} \leq 1\}.$$

Main observation: one can relate solutions of

$$\prod_{i=1}^k n_i = \prod_{j=1}^k m_j$$

to arrays that are very similar to those that appear when studying c_n .

Informally: suppose we are given positive integers $(R_{i,j})_{1 \leq i,j \leq k}$. Define

$$n_i = \prod_{j=1}^k R_{i,j}, \quad m_j = \prod_{i=1}^k R_{i,j}.$$

Then

$$\prod_{i=1}^k n_i = \prod_{j=1}^k m_j = \prod_{1 \leq i,j \leq k} R_{i,j}.$$

Still continuing informally, we may write

$$\sum_{\substack{n_1, \dots, n_k \leq x \\ m_1, \dots, m_k \leq x \\ \prod n_i = \prod m_j}} \frac{1}{\sqrt{n_1 \cdots n_k}} \frac{1}{\sqrt{m_1 \cdots m_k}} \approx \sum_{\substack{\prod_{j=1}^k R_{i,j} \leq x \\ \prod_{i=1}^k R_{i,j} \leq x}} \frac{1}{\prod_{1 \leq i, j \leq k} R_{i,j}}.$$

Note that these conditions are multiplicative analogues of the arrays counted in Diaconis–Gamburd.

By a standard real-analytic argument, this will grow like $\text{vol}(\mathcal{P}_k)(\log x)^{k^2}$. Where is a_k ?

Main issue: the map that takes the $k \times k$ array $(R_{i,j})_{i,j \leq k}$ to the pair of tuples $(\prod_{j=1}^k R_{i,j})_{i=1}^k$ and $(\prod_{i=1}^k R_{i,j})_{j=1}^k$ is not a bijection, because it is not injective.

Define

$$\text{Co}_k = \{R \in M_k(\mathbb{N}) : \gcd(R_{p,q}, R_{r,s}) = 1 \text{ when } p < r \text{ and } q > s\}.$$

Lemma 1 (Wooley–Vaughan, Montgomery–Soundararajan)

The map

$$(R_{i,j})_{i,j \leq k} \mapsto \left(\left(\prod_{j=1}^k R_{i,j} \right)_{i=1}^k, \left(\prod_{i=1}^k R_{i,j} \right)_{j=1}^k \right)$$

is a bijection from Co_k onto the set of pairs of tuples $(n_1, \dots, n_k)$ and $(m_1, \dots, m_k)$ with $\prod_{i=1}^k n_i = \prod_{j=1}^k m_j$.

This leads to the following rigorous equality:

$$\sum_{\substack{n_1, \dots, n_k \leq x \\ m_1, \dots, m_k \leq x \\ \prod n_i = \prod m_j}} \frac{1}{\sqrt{n_1 \cdots n_k}} \frac{1}{\sqrt{m_1 \cdots m_k}} = \sum_{\substack{\prod_{j=1}^k R_{i,j} \leq x \\ \prod_{i=1}^k R_{i,j} \leq x \\ (R_{i,j})_{1 \leq i,j \leq k} \in \text{Co}_k}} \frac{1}{\prod_{1 \leq i,j \leq k} R_{i,j}}.$$

How to analyze? Write as

$$\sum_{(R_{i,j})_{1 \leq i,j \leq k} \in \text{Co}_k} \frac{1}{\prod_{1 \leq i,j \leq k} R_{i,j}} f \left(\left(\frac{\log R_{i,j}}{\log x} \right)_{i,j} \right)$$

where f is the indicator of the $\mathcal{P}_k$ that appeared before.

Tauberian trick: if one can analyze this when $f(\vec{s}) = e^{-\vec{s} \cdot \vec{\lambda}}$, then one can analyze this for indicators of convex, bounded sets.

Moments of $\sum_{n \leq N} c_n$ are related to arrays where row and column sums are bounded by N . What about moments of $\sum_{n=0}^N (-1)^n \text{Sc}_n(U_N)$?

It is natural to ask, because

$$\sum_{n=0}^N (-1)^n \text{Sc}_n(U_N) = \det(I - U_N)$$

is analogous to

$$\zeta(1/2 + it).$$

Given a $k \times k$ array A of nonnegative integers, a South–East chain is a sequence of distinct entries of A in which each entry is located weakly to the right of and weakly below the preceding entry.

The length of a chain is the sum of entries in the chain.

We define

$$\text{SE}(A)$$

as the maximal length of a South–East chain in A .

Theorem 3 (Baik–Rains, 2001)

The expectation

$$\mathbb{E}\left[\prod_{i=1}^k \text{Sc}_{n_i}(U_N) \prod_{j=1}^k \overline{\text{Sc}_{m_j}(U_N)}\right]$$

equals the number of $k \times k$ arrays A of nonnegative integers, where the i th row sums to n_i , the j th column sums to m_j , and additionally

$$\text{SE}(A) \leq N.$$

Consequently,

$$\mathbb{E}|\det(I - U_N)|^{2k}$$

equals the number of $k \times k$ arrays A of nonnegative integers, where the row sums and column sums are bounded by N , and additionally $\text{SE}(A) \leq N$.

When studying c_N , we encountered arrays with row-sum and column-sum constraints.

When studying $\alpha(n)$, we encountered arrays with row-product and column-product constraints.

This suggests defining the *multiplicative* length of a chain is the product of the entries in the chain, and

$$\mathrm{SE}^\times(R)$$

as the maximal *multiplicative* length of a South–East chain, and

The analogy of U_N and ζ suggests that

$$\frac{1}{T} \int_0^T \left| \zeta \left(\frac{1}{2} + it \right) \right|^{2k} \sim \sum_{\substack{R_{i,j} \in \text{Co}_k \\ \text{SE}^\times(R) \leq T}} \frac{1}{\prod_{1 \leq i,j \leq k} R_{i,j}}.$$

We are able to prove that this is consistent with the Keating–Snaith prediction, again using a Tauberian trick.

Theorem 4 (Gorodetsky–Rodgers)

$$\sum_{\substack{R_{i,j} \in \text{Co}_k \\ \text{SE}^\times(R) \leq T}} \frac{1}{\prod_{1 \leq i,j \leq k} R_{i,j}} \sim a_k g_k (\log T)^{k^2}.$$

Moreover, $g_k = \text{vol}(\mathcal{Q}_k)$ where

$$\mathcal{Q}_k = \{(x_{i,j}) \in \mathbb{R}_{\geq 0}^{k^2} : \sum_{i=1}^k x_{i,j} \leq 1, \sum_{j=1}^k x_{i,j} \leq 1, \text{SE}(x) \leq 1\}.$$

We hope that this result will shed light on moments of zeta. Yet to be seen.

We wish to briefly explain the result of Baik–Rains. It relies on symmetric function theory.

Schur polynomials $s_\lambda(x_1, \dots, x_N)$ are indexed by partitions. They are important in the study of U_N because when x_i are replaced with eigenvalues of U_N , we obtain irreducible characters of U_N , and in particular

$$\mathbb{E} \left[s_\lambda(U_N) \overline{s_\mu(U_N)} \right] = \delta_{\lambda, \mu} \mathbf{1}_{\ell(\lambda) \leq N}.$$

There are many bases of symmetric polynomials, and we know how to pass between them.

For example: $\text{Tr}(U_N^k)$ is the k -power sum symmetric polynomial in the eigenvalues of U_N , denoted p_k . By expressing p_k and its powers as a linear combination of Schur polynomials, Diaconis and Shahshahani proved that

$$\mathbb{E}|\text{Tr}(U_N^k)|^{2m} = k^m m!$$

holds for $mk \leq N$, hence $\text{Tr}(U_N^k)/\sqrt{k} \xrightarrow[N \rightarrow \infty]{d} N_{\mathbb{C}}(0, 1)$ when $k = o(N)$.

This is highly relevant to the study of secular coefficients: $Sc_n(U_N)$ is the n th elementary symmetric polynomial in the eigenvalues of U_N , denoted $e_n(U_N)$. We have

$$\prod_{i=1}^k Sc_{n_i}(U_N) = e_{\mathbf{n}}(U_N)$$

where $\mathbf{n} = (n_1, \dots, n_k)$ and $e_{\mathbf{n}} := \prod_i e_{n_i}$.

Fact 1: if $|\lambda| = n$ then

$$e_\lambda = \sum_{\mu \vdash n} K_{\mu', \lambda} s_\mu.$$

where the sum is over partitions of n , μ' is the conjugate partition, and K is the Kostka number which counts *semistandard Young tableaux of shape μ' and content λ* .

Thus, if $\sum_{i=1}^k n_i = \sum_{j=1}^k m_j = n$ then

$$\mathbb{E}\left[\prod_{i=1}^k \text{Sc}_{n_i}(U_N) \prod_{j=1}^k \overline{\text{Sc}_{m_j}(U_N)}\right] = \sum_{\substack{\mu \vdash n \\ \ell(\mu) \leq N}} K_{\mu', \mathbf{n}} K_{\mu', \mathbf{m}} = \sum_{\substack{\mu \vdash n \\ \mu_1 \leq N}} K_{\mu, \mathbf{n}} K_{\mu, \mathbf{m}}.$$

Fact 2: The RSK algorithm gives a bijection between pairs of semistandard Young tableaux of equal shape, and arrays of nonnegative integers.

The sums of rows and columns in the array are dictated by the content.

The longest SE-chain of the resulting array is equal to the largest part in the shape.

Thank you for listening.